

Route lifetime: cycle vs event trigger, by planner

With event triggering a route is kept 3.2–4.7× longer at 20 ms checks (Voronoi 110 → 518 ms). Average time a route is followed before a new one replaces it. Cycle calls the planner at every check, but while the straight line to the target is clear the robot keeps driving straight and no new route is made; only a detour (or a step out of an obstacle zone) replaces the route. Cycle routes therefore outlast one check, and long straight-line stretches raise every average.

